

1) Incident Overview (Executive Summary)

During routine security monitoring using **Elastic SIEM**, multiple suspicious events were detected, including malware alerts and repeated authentication failures. The activity was identified through log analysis in the **Discover module** and visualized using the **Security Monitoring Dashboard**. The correlation of these events indicates potential system compromise and unauthorized access attempts. Due to the severity of the alerts, immediate investigation and mitigation are required to prevent further impact.

2)Detection Method

The incident was detected using Elastic Stack (ELK) SIEM. System and security logs were ingested into Elasticsearch and analyzed using the Discover module. Keyword-based filtering was applied to identify suspicious events such as malware detections and failed login attempts. Additionally, a SIEM dashboard was created using Elastic Lens to visualize security alerts.

3)Scope of the Incident

The incident affected multiple user accounts and systems within the monitored environment. Logs indicate activity associated with several internal IP addresses and user accounts, suggesting that the suspicious behavior was not isolated to a single system.

4) Alert Classification

ALERT ID	DATE & TIME	USER	IP ADDRESS	EVENT TYPE	THREAT	SEVERITY
A1	2025-07-03 05:48:14	BOB	10.0.0.5	Malware detected	Trojan	HIGH
A2	2025-07-03 04:19:14	ALICE	198.51.100.42	Malware detected	Rootkit	HIGH
A3	2025-07-03 09:10:14	BOB	172.16.0.3	Malware detected	Ransom ware	HIGH

A4	2025-07-03 07:02:14	ALICE	203.0.113 .77	Login Failed	—	MEDIUM
A5	2025-07-03 06:01:14	BOB	172.16.0. 3	File accessed	—	LOW

5)Incident Timeline

TIME	EVENT DESCRIPTION
04:19	Rootkit malware detected on a user system(primary incident)
05:48	Additional malware activity (Trojan) observed on another system
07:02	Multiple failed login attempts detected
09:10	Ransomware-related behavior identified

6)Indicators of Compromise (IOCs)

Based on the analysis of security logs in Elastic SIEM, the following indicators of compromise were identified during the investigation:

Malware Indicators

- **Trojan malware detections**, indicating the presence of malicious software capable of unauthorized access and data compromise
 - **Rootkit signature detection**, suggesting stealthy persistence mechanisms on affected systems
 - **Ransomware-related behavior**, indicating potential encryption or extortion activity
 - **Worm infection attempt**, pointing to possible self-propagating malware behavior
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Authentication & Account Indicators

- Multiple **failed login attempts**, which may indicate brute-force or credential-stuffing attacks
 - Login failures followed by successful authentication events, suggesting potential **account compromise**
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Network & Host Indicators

- Repeated activity from specific internal and external IP addresses associated with suspicious behavior
 - File access activity observed in proximity to malware detection events, which may indicate **post-compromise activity**
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7) Impact Assessment

The discovered security events point to the possibility of vulnerability of the targeted systems and accounts. The discovery of Trojan, Rootkit, and Ransomware behavior with respect to malware represents serious threats to the integrity, confidentiality, and availability of the targeted systems. The threats, if not addressed, have the capability of allowing unauthorized access, stealing sensitive data, disrupting, or encrypting vital files.

Moreover, the identified authentication errors could well represent attempts at credential theft, further adding to the risks of unauthorized access to the internal infrastructure. Despite the absence of any evidence related to data exfiltration, there remains ample concerns related to the security implications of the malware and login errors identified above.

8) Response & Mitigation Actions

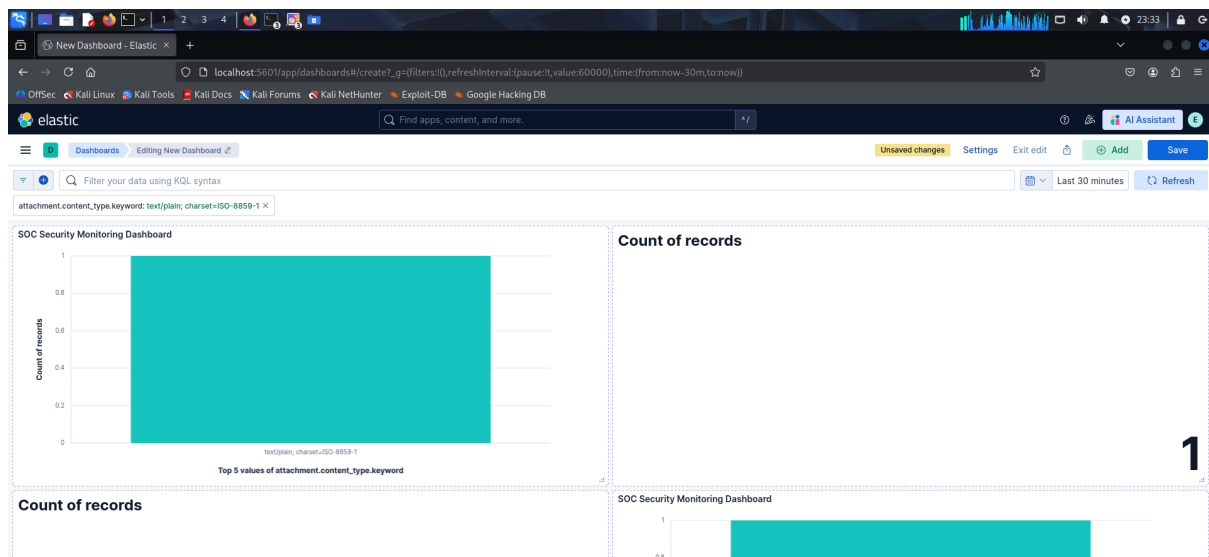
On detection of the security incident, the following measures shall be taken:

- Isolation of the affected systems to contain the spreading of the malwares.
- Performing comprehensive system and endpoint security scans to detect and eliminate harmful files.
- Reset credentials for affected user accounts and use stronger password policies.
- Monitor & block malicious IP addresses associated with malicious & suspicious activity.
- Check system and security event logs for other indicators of compromise.
- Enhance the triggering conditions of SIEM alerts to better identify similar threats in the future.
- Make sure that all systems have the most recent security updates installed.

These steps are taken to control this incident, avoid future occurrences to a large extent, and improve the security posture of the organization.

9) Dashboard & Monitoring Evidence

A SIEM dashboard was created using Elastic Lens to monitor failed login attempts and malware detections. The dashboard provides high-level visibility into ongoing security events and supports continuous monitoring and alert triage.



10) CONCLUSION

In this scenario, it is clear that elastic security information and event management with the help of the Elastic SIEM system played an important role in allowing several security incidents, such as malware and authentication anomalies, to be discovered and analyzed. The key aspect in this scenario is the early detection that allows risks to be assessed.

Through the focus on high alert messages and the correlation of associated activities, the investigation offered a clear view of the nature and potential of the incident that occurred. Adoption of the proposed mitigation plans and improvement in the way the organization is monitored will help to prevent similar incidents from happening in the future.